

Unlocking the Tacit Knowledge of Data Work in Machine Learning

Inha Cha*
inhacha@upstage.ai
Upstage AI Research
Yongin, Korea, Republic of

Juhyun Oh*
juhyun@upstage.ai
Upstage AI Research
Yongin, Korea, Republic of

Cheul Young Park*
cheul@upstage.ai
Upstage AI Research
Yongin, Korea, Republic of

Jiyeon Han
jiyeonhan@upstage.ai
Upstage AI Research
Yongin, Korea, Republic of

Hwalsuk Lee
hwalsuk.lee@upstage.ai
Upstage AI Research
Yongin, Korea, Republic of

ABSTRACT

Creating datasets for ML is an inherently human endeavor, as the data's heterogeneity mandates human intervention. However, most data workflows being one-time and hardly transferable leads to a lack of standardization and reusability. There has been a push to impose more structure on the data work process, but little is known about the implicit or "tacit" knowledge of data workers, i.e., "know-how"s that is difficult to transfer to others. Identifying and formalizing this knowledge can help data work improve, leading it from current "exploration" to more systematic "engineering." We interviewed 19 ML practitioners in this study to find "why" they use "what" tacit knowledge. As a result, we identified the following themes: 1) data is context/situation dependent, 2) human workers are inseparable from data, and 3) models must be understood to build data. We finally discuss future systematic supports and research to convert what is implicit to explicit.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Data Construction, Machine Learning, Practitioners, Semi-structured In-depth Interviews

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1 INTRODUCTION

Machine Learning is advancing rapidly, but the data construction process: from data design, collection, and preprocessing, to model

training and evaluation, i.e., the *data work* as a whole and the reiteration of it, is still primarily a human endeavor. The diversity and complexity of data and associated issues make humans inseparable from data work. However, a greater problem lies in that most data workflows are one-time and lack transferability [27]. For example, scripts or notebooks used for data preprocessing are rarely reused after their initial use [14, 32]. This is in contrast to the importance placed on high reusability in almost all programming practices, including ML modeling and training. This phenomenon of the outputs of data work being only temporary is prevalent throughout data work.

Data work for ML produces not only tangible outcomes like preprocessing codes and the data itself but also intangible outcomes such as the rationale for evaluating data quality and strategies for maintaining data consistency and addressing edge cases. However, most of this knowledge is temporary and only used during a single data workflow and is often forgotten or limited to the personal knowledge of a data scientist or a team's tribal knowledge. While this "exploratory" attitude that dominates the data community is useful for "wrangling" with a variety of data [32], it results in people doing data work relying on trial-and-error to create solutions as needed without establishing a systematic approach to create quality data efficiently.

Many studies have attempted to improve the situation by imposing more structure on data work. For example, some studies advocate for more rigorous documentation [6], while others propose visualization tools that support tracking data scientists' work history and internal discussion among teams [13, 29]. There are also works that attempt to establish a theory of data science work based on qualitatively observing data scientists and their work [16, 32]. The common goal of these works is to make what is implicit explicit. However, there has been little examination of the implicit knowledge, or tacit knowledge [17], that data workers acquire and use in data workflows.

Tacit knowledge has been studied in various domains, such as knowledge management and organizational behavior, and is described as "know-how" or "knowing how to do something [7, 18, 26]." It is distinct from domain knowledge, which is knowledge about a specific field of expertise and can be learned through formal education and training. Tacit knowledge, on the other hand, is acquired through experience, practice, or observation and is difficult to transfer to others [21].

*These authors contributed equally to this work.

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In machine learning, efficiently creating high-quality data may require an understanding of a specific subject domain, as well as machine learning domain knowledge. However, tacit knowledge, "knowing-how" to communicate with annotators, make trade-offs in the face of conflicting interests, and build consistent data; those strategies/rationales/intuitions gained via experiences are mandatory for a successful data project. Thus, identifying and formalizing this tacit knowledge can help data work improve, leading it to head from current unstructured "exploration" to more systematic "engineering" [2, 11] and improve the quality and accountability of resulting data [19].

Towards this goal, we aimed to understand what tacit knowledge people use in data work and why they use it in this study. To do this, we interviewed 19 practitioners on how they build datasets for ML projects and analyzed the results through iterative thematic coding. We discovered that individuals utilize distinct tacit knowledge depending on their contexts, but for three common reasons: (1) *data is dependent on context and situation*, (2) *human involvement is inseparable from creating data*, and (3) *decisions about data must be made based on an understanding of the model*. With these findings, we discuss how certain tacit knowledge can be addressed through systematic support and where further research is needed to convert tacit knowledge into explicit knowledge.

2 RELATED WORKS

2.1 Human influences in data work

Humans play a significant role in data work. For example, they determine whether and how a dataset might be useful for a specific application, create data firsthand, conduct thorough error analysis, and handle the numerous unstandardized difficulties of creating datasets [10, 20, 32]. In that regard, understanding human engagement in constructing datasets for ML has risen in interest among HCI researchers [19]. In particular, how humans engage with data; what considerations they take into account during data work; biases in their judgments; volatile circumstances arising from human involvement are a few topics that the community is actively investigating.

This effort is distinct from data-related works in ML [12, 23, 24], as they focus on data engineering in ML workflows and primarily introduce the technical skills required for specific steps in data preparation (e.g., data collection, cleansing, and validation) rather than human influences. On the other hand, much work in the field of HCI is bridging the gap between the technical aspect of building ML datasets and understanding the roles and influences of humans making ML data [3, 5, 28, 30, 32]. Previous HCI research provides a holistic perspective on how people do data works [16, 32] while also delving into more nuanced topics regarding human's role in data work, such as the power dynamics of various stakeholders in annotation work [15], how humans may improve the fairness of ML systems [9], data problems caused by humans and their potential resolution [25], disparate values data workers with different roles assigned to data [28], and how humans collectively craft ground truth of data [17]. In line with these works, our study shines a light on how and why humans in data work use tacit knowledge in practice.

2.2 Implicit knowledge in data work

Previous works have reported that, on average, 80% of the entire time in a data science project is spent on data preparation [8, 16, 22, 34], where much manual effort is needed. They also pointed out that among the whole ML workflow, steps related to data work require most human interaction and domain expertise, resulting in communication bottlenecks. Researchers struggle to automate such "data wrangling" steps because human judgment and engagement are mandatory in data processes [12, 31]. This tight dependency between humans and data makes data work iterative and thus time-consuming.

Nonetheless, even before considering automation, parts that require human engagement in data work are challenging to structure or transfer to others due to the abstract and implicit nature of knowledge we employ — namely the tacit knowledge, domain knowledge (i.e., knowledge about the subject domain where the ML model is being applied), in addition to the prior experiences and intuitions. For example, [16] stated that data science is currently understood as an exploration process that necessitates implicit knowledge. [31] also reported that humans use contextual awareness and domain knowledge to collect, label, and clean data in ML workflow.

Among such hard-to-articulate types of knowledge, tacit knowledge is particularly difficult to formalize, requires contextual knowledge, is personal, and is implicit [7, 18, 26]. Even those who use tacit knowledge may be unaware of the fact that they are using such knowledge. [16, 17] noted that the very nature of knowledge used in data work is to be tacit. However, little attention has been paid to the use of tacit knowledge in data work. Therefore, in this work, we focus on tacit knowledge used in data work and its characteristics. In doing so, we investigate how to transform this implicit knowledge into controllable, expectable, and principled knowledge.

3 METHODS

3.1 Recruitment and Participants

The participants were recruited from the authors' personal networks, online developer communities, and within the authors' affiliated organization (a Seoul-based AI tech startup) via snowball and purposive sampling. As a result, 19 individuals affiliated with 13 different organizations participated in the study. Since the authors were interested in people's use of tacit knowledge for constructing ML datasets, only those with first-hand experience building data to deploy real-world ML applications or extensive research were invited to participate.

The participants were located in or had worked on projects based in South Korea or the United States. They had a minimum of 2 to a maximum of 13 years of experience in ML, with a median of 7 years, and their occupations varied from AI Research Engineer, ML Engineer, SW Engineer, Data Scientist, Ph.D. Student, and Data Manager. Participants' domains of expertise also varied, with the most common being Computer Vision (11 participants) and subdomains ranging from OCR, bioinformatics, medical, to multimodal, then Natural Language Processing (6 participants), and 2 in Music Information Retrieval. Refer to the Table 1 for detailed information on participants.

Table 1: Columns from the left: (1) participant #, (2) job/role, (3) years of experience in programming + ML, (4) domain, (5) previous project.

No.	Job/Role	Yrs. of Exp.	Domain	Previous Proj.
P1	AI Research Engineer	6	NLP	Opinion mining from product reviews
P2	AI Research Engineer	5	CV, Multimodal	Physical activity recognition
P3	ML Engineer	7	CV, OCR	Checkbox recognition from documents
P4	Data Manager	2	CV, OCR	Medical documents parsing
P5	AI Research Engineer	8	CV	Detecting defects in manufactured products
P6	AI Research Engineer	4	NLP, Speech	Automatic speech recognition
P7	AI Research Engineer	11	OCR	Medical documents parsing
P8	AI Research Engineer	7.5	CV, Bioinformatics	Detecting defects in manufactured products
P9	AI Research Engineer	4	CV, Medical	Detecting tables in documents
P10	Software Engineer	13	NLP	Neural Machine Translation
P11	Data Scientist	7	CV	Fashion image search and recommendation
P12	AI Research Engineer	6	CV	Defect detection for mobile phones
P13	Ph.D. Student	9	MIR	Automatic DJ/mixing
P14	Software Engineer	3	OCR, CV	Credit card OCR
P15	Ph.D. Student	4	Speech, MIR	Singing Voice Synthesis
P16	Ph.D. Student	8	NLP	Neural Machine Translation
P17	Ph.D. Student	5	CV	Object detection for classifying mart goods
P18	AI Research Engineer	3	NLP	Hate speech detection
P19	AI Research Engineer	2	NLP	Sentiment analysis from movie reviews

3.2 Interview Procedure

Interviews were conducted online, and all sessions were recorded, transcribed, and anonymized with the participant’s permission. The interviews were semi-structured, and we asked participants to talk about their experience collecting data to build an ML application or conduct a research project. At the beginning of the interviews, participants were informed that they could discontinue without disadvantage at any time if they found it uncomfortable to continue participating in the research for any reason.

In each interview session, the interviewers briefly explained the research goal of investigating data work practices in Machine Learning, not limited to data labeling but covering range of tasks related to data in a full ML life-cycle, and asked the participant about their professional background and experience in building a dataset for AI/ML. In particular, we instructed participants to choose one specific project in which they built a dataset and used it to deploy an ML model for production or conduct research. Then, on the topic of their choice, participants were told to provide a detailed account of the data construction steps from the beginning to the end, including iterations, giving each step a descriptive name such as "Data Design" or "Data Preprocessing," while focusing on the difficulties of data construction in each step and their measures and know-hows to overcome those. Finally, all participants were financially compensated the amount of 50,000 KRW for participation.

3.3 Qualitative Analysis

From the transcribed interviews, the authors performed thematic coding using Dovetail to extract themes related to the use of tacit knowledge during data construction for AI/ML. Initially, each author read the transcriptions multiple times individually, looking for common themes that grouped quotes and assigning descriptive codes, such as "difficulty in maintaining consistency in data." Then, through iterative discussion, the authors constructed upward from basic themes to refined, higher-level themes explaining why tacit knowledge is necessary during data construction. As a result, three major themes were identified, along with multiple sub-level themes underneath.

4 FINDINGS

In this section, we present findings from interviews with 19 machine learning experts and a thematic analysis of their responses. We found that people use various tacit knowledge depending on specific contexts, making it difficult to generalize. Nonetheless, they use tacit knowledge due to the following three major reasons: 1) the context dependency of data, 2) the costliness of human data workers, and 3) the inscrutability of deep neural networks. These themes are not mutually exclusive but are interconnected with each other to explain why tacit knowledge are mandatory for data construction. We also organize our findings under six steps of data construction identified in the interviews: 1) Data Design [DD], 2) Data Exploration & Collection [DC], 3) Data Annotation [DA], 4)

Data Preparation [DP], 5) Data Evaluation & Refinement [DE], and 6) Model Deployment [MD].

4.1 Data is dependent on context and situation

In theory, "good" data is large in quantity, represents the population, accurately describes reality, includes edge cases, is maintainable, and ultimately leads to the best model performance [4]. However, in practice, data must consider various constraints depending on the context, and data workers must make the best choices among various trade-offs to achieve their goals. Participants stated that the proper strategy for building data in conflicting situations, i.e., tacit knowledge, is necessary.

4.1.1 Ground-truth labels are designed or created per the situation [DD, DA]. Most participants knew that it is impossible to create data with a ground truth that encompasses all situations. P5 shared his experience on how ground truth is "made" in situations where the interests of multiple stakeholders must be taken into account. *"After discussing with the client, we just grouped things that looked ambiguous into one class. [...] if we didn't know whether something is normal or a defect, then it was better to just classify them as either of the classes."* Other participants also stated that it is vital to consider the model's learning capacity when defining the appropriate level of granularity for ground truth labels.

P11 discussed the experience of building a fashion recommendation service by defining made-up "style labels" and labeling clothes images with those labels to achieve the business agenda of giving high exposure to clothes with specific styles. *"For example, this 'style' doesn't exist as a ground truth anywhere, so it has to be fabricated. [...] There is this fashion team who determines such categorizations. So, if the fashion team decides on the styles they want the customers to be exposed to, I then go figure out how to collect data based on that."*

As the data itself is ambiguous and does not have an absolute truth, participants had to make up a subjective ground truth appropriate for their respective purposes. Therefore, knowing the necessity of inventing and adapting "ground truth" and applying it appropriately depending on the situation can be considered tacit knowledge.

4.1.2 "Consistent" data is difficult to make [DA]. Consistency is critical to training data for deep learning models: it has to be without conflicting information that would harm the model's learning. However, consistency is mostly defined by people in a context-dependent manner. Participants said that creating perfectly consistent data is almost impossible — that consistency and how well data captures reality compensate each other, so it is necessary to adjust the consistency of the data depending on the situation.

For example, P4 and P5 mentioned this dilemma of creating consistent labels. They said that if strict and mechanical criteria are given in the annotation guide for high consistency, the data may not properly reflect reality, and if the criteria are too loose and left to the workers' discretion, the consistency of the data may be lost. To cope with this dilemma, participants concocted solutions from their own experiences. P4 noted: *"in my opinion, consistency is a must when you try to do something with data. The data can be wrong,*

and that's fine. Of course, it may vary from person to person, but I'd rather have my data consistent than have many correct answers."

In case of P14, depending on the type of data and the ability of the annotators, consistent data could be built by assigning the task to one annotator, but also by having multiple annotators derive consensus. Also, P6 and P12 emphasized gathering exceptions and edge cases to create consistent data. They described reflecting well-documented exceptions and edge cases into the annotation guidelines as often as possible as their go-to strategy.

4.2 Humans are inseparable from data

Data work inevitably requires human involvement. However, humans act as a very constraint on data work. In that regard, our participants used tacit knowledge in understanding and managing people involved in data construction, particularly the annotation step.

4.2.1 Human annotation imposes quantity-quality tradeoff [DD, DA]. Human engagement in data work leads to quantity (speed) vs. quality tradeoffs. In some situations, it is necessary to prioritize speed and collect data in large quantities, while in other cases, quality data is mandatory, even if it is a handful. Participants said that, ultimately, it is essential to decide what to compromise in respectful situations and have the *know-how* to make that decision. For example, P17 commented – *"Now, in the case of a supermarket, there were tens and hundreds of types of snacks, and it was very slow and tedious for labelers to find and label each of them one by one. So while data needed to be collected quickly and in large quantities, it took a long time to make one image, so the quality of each sample was not up to par."* In such situations where there is no clear cut solution, participants had to base their judgments on their previous experiences.

4.2.2 The costliness of human annotation mandates a cost-effective approach from the beginning [DD]. As with previous studies [17, 30, 33], our participants also noted that data construction is costly, as it involves people. Therefore, practitioners prioritized cost-effectiveness from the very first data design. Participants said that it is imperative to "strategically" create expandable data because re-designing and collecting data anew is expensive. P1, in particular, highlighted the reusability of the labeling schema that can also adapt to changes in a model architecture. Furthermore, P1 and P19 emphasized the importance of building data with scalability across subject domains in mind and checking whether the current data format can also be responsive to incoming data upon the deployment of the model for service.

4.2.3 Understanding (of) annotators determines the quality of the data [DA]. Participants commonly agreed that the quality of annotations is determined in the annotator's hands. In particular, P6 stressed that *"the caliber of annotators"* is crucial to obtaining quality data. However, participants also noted that the limits in the abilities of individual annotators and the discrepancy in them also impose an upper bound on the quality of data. P7 used a strategy of clarifying areas where the annotator must not make mistakes and where the annotator can use their judgment to increase labeling quality to help them understand the elements most important for the task.

Participants also emphasized the importance of adjusting the annotators' workload. For example, P17 commented it is important that managers monitor the speed at which an annotator completes one image to determine the suitable workload for a day. At the same time, P2, P11, and P12 stressed the importance of allocating appropriate break time to prevent annotators from becoming fatigued. Participants also noted that understanding the annotators' motivations and working conditions is especially important in annotation as it is often tedious. In that regard, our participants found adjusting the difficulty of tasks according to the annotator's capacity [P19], explaining to annotators the ultimate "why" of creating data, and emphasizing the impact that data quality can have on modeling can help annotators be more motivated [P7, 19]. P7 recollected: "[...] people are not robots, so give them the big picture showing what we are trying to create with the data. Tell them (annotators) how we will use the data to help many people reduce their time on repetitive tasks. [...] We can also ask them to imagine what the data should look like to achieve that goal. After we gave them training, some annotators even began to report on edge cases. [...] So the lesson learned was how important it is to educate them and, eventually, inspire them."

4.3 Data decisions require model understanding

Data design, evaluation, and other data-related decisions must be made with a sufficient understanding of models. Currently, most AI models are based on deep learning. However, our lack of understanding of this technology and the current limitations of deep learning impact data decisions. P6 said, *"I have thought about it, but I could not come up with an answer. It is actually the desire of all people (data scientists). How to choose and add the data, so the model's performance improves."* Participants used tacit knowledge to infer the cause and effect between the model and data when they could not understand the precise causality between them.

4.3.1 Designing data and annotation schema needs model results [DD, DA, DE]. Participants stated that understanding how the model operates and the model's results have a significant impact on determining the annotation task and schema. They said performance does not necessarily improve by unconditionally increasing the amount and diversity of data. Hence, the detailed relationship between the annotation system and model performance always needs empirical investigation. P14 commented, *"How can we find the parts where the model might be confused and how to make it clear? It's all experience. Someone who has done a lot (of modeling) will have rough insights into a model and know what to do to avoid baffling a model."*

When making language processing (NLP) models, P1 had to decide what data to prepare and how to formulate tasks based on the model predictions, as it was difficult to determine the precise number and data schema from the beginning without accurate knowledge of the model's limitations. P3, P9, P15, and P16 also reported quickly testing model performance with a small amount of data at the beginning of their projects and then slowly adjusting annotation schemes and supplementing guidelines in a direction that improves model performance. Especially, P9 and P16 mentioned that it is essential to keep checking the deltas in model performance and slowly increase the amount of data while tuning the filtering threshold to add or remove data noise for model robustness.

4.3.2 Data evaluation also depends on model outputs [DE, MD]. P14 reported that analyzing model failure cases helps build datasets, and the process relies on empirical intuition to identify such cases. P14 said: *"The model's failure may be because the model architecture is built wrong, the distribution of training data is unsuitable to tackle the problem, the data is misleading due to incorrect labeling, or else, there is no way of knowing without examining each image one at a time."*

On the other hand, P4 states that in some cases, the quantitatively measured performance of a model does not align with human assessment, and experts use intuition to make amendments even if a direct causal relationship is not apparent: *"[...] even if the model metric reads 98%, it doesn't always live up to the expectation in production. There's usually a difference between how a person feels and what the number says. [...] When we (data managers) see this (discrepancy), we know where to look for – the test set or the model."*

5 DISCUSSION AND CONCLUSION

Our findings provide insight into the contexts of data work tacit knowledge are used and what leads to the use of tacit knowledge. Based on these findings, we discuss opportunities to convert implicit knowledge to explicit knowledge in light of three themes from our interview: data, humans, and models.

5.1 Scaffolding data workflows to guide decision making

The need for data arises in diverse situations and involves the interests of multiple stakeholders and even business objectives. Therefore, it would be nearly impossible to automate decision-making for data work without considering all variables in a given situation. However, for those needing help from data experts, guidance to navigate data workflows and support decision-making in situations where value conflicts arise could be provided systematically. Scaffolding diverse scenarios and providing advice on decisions can help even those without data experience benefit from expert knowledge. Tools like Apache Airflow [1], which are used for workflow authoring and monitoring, are commonly used for data ETL (Extract, Transform, Load) processes. Similarly, a management tool for data workflows can be developed. In that case, it could allow data experts to document their workflows and the underlying rationale for each decision in detail [29], which can be shared with the community to standardize data workflows collectively.

Further, if a system could inform data scientists of the progress of annotation tasks in real-time, whether through statistics or visualization, they would be able to make more informed decisions during the data annotation step. For example, if the managers can see the patterns of annotators' labeling, such as which tags are not being used or which tags are misunderstood, it would be easier for them to control consistency in annotation works. Currently, most of this information is obtained manually. But suppose a system can provide interactive support highlighting potentially faulty parts of ongoing annotation tasks. In that case, managers could intervene immediately, which would help to improve consistency among different annotators in various working scenarios.

5.2 Assessing and managing human annotators' competence and motivation

Assessing and managing the abilities of human annotators is crucial in data work. In that regard, many aspects of annotator management that are currently implicit could be made explicit to improve the state of data work. First, labeling quality is affected by the "competence" of human annotators; thus, systematic support could be provided to measure it quantitatively. Factors such as the annotator's work speed, consistency with other workers' labels, and understanding of the task's end goal are crucial components of the annotator's ability. However, these factors are not explicitly measured or monitored in most current annotation systems. Even if evaluation metrics like Inter-Annotator Agreement exist, utilizing them without systems providing internal support is not easy. If these quantitative metrics can be well utilized, it will be possible to accumulate and share explicit knowledge about annotation work based on various experiments and research, i.e., which results from workers with certain qualities are more appropriate in each situation, rather than just expressing it ambiguously as "work sense."

On the other hand, it is also important to move away from treating workers as simply "resources" and to make efforts to consider their motivations and careers as "human." [30] From the perspective of datasets as infrastructure [11, 19], converting data workers' tacit knowledge about the "human" aspect into something explicit and replicable will require accumulating collaboration practices among multiple stakeholder groups. For example, like P4's organization, it may be possible to separate data managers who directly create and manage data from ML engineers.

5.3 Overcoming the ambiguous causality between models and data

Experts use tacit knowledge to overcome the unclear causal relationship between models and data. The participants in our study suggest the need for research to increase the interpretability of AI. To make the tacit intuition of ML and data experts explainable, the ML and data community may develop novel metrics providing more insight into models, data, and their relationship. First, intuitive metrics that align with both human evaluations and model performance are needed. For example, P9 and P16 stated that even if the model's performance score is high, the results may not be convincing to a human. Research on score calibration based on confidence scores is needed to find the model's weak spots. To fully understand the relationship between models and data, quantitative metrics for evaluating the data itself are necessary, rather than just relying on goodness-of-fit. Research is needed on what constitutes "goodness-of-data" [4].

However, for these metrics to be adopted into data workflow, they must be accompanied by a system ensuring that judgments based on the metrics can be smoothly reflected in the next iteration. In addition, it must be easy to analyze which parts of data have been improved and what results were obtained by removing or adding certain data. For example, it is possible to visualize the differences between data versions or to assist in identifying data differences algorithmically, as in [27].

5.4 Conclusion

In this work, we found that tacit knowledge in data work is not ought to remain tacit. As the importance of data in ML increases, we call for efforts to make data work more disciplined that has value in itself, rather than being viewed as tedious, time-consuming, and unstructured, by breaking out of current stereotypes. This includes taking a more systematic, engineering-like approach toward data work, incorporating more rigorous documentation practices, and actively investigating opportunities for further studies.

A limitation of our study is the small sample size ($n=19$) which might not be representative of data workers in general. In addition, the scope of our work is limited to works done by experienced data scientists (data managers and ML engineers), without considering data annotators who play crucial role in data work for ML by labeling and structuring the data. Future work may cover more diverse stakeholders and contributors of data work, including data annotators and participants outside the technology industry, to confirm and expand upon our findings.

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